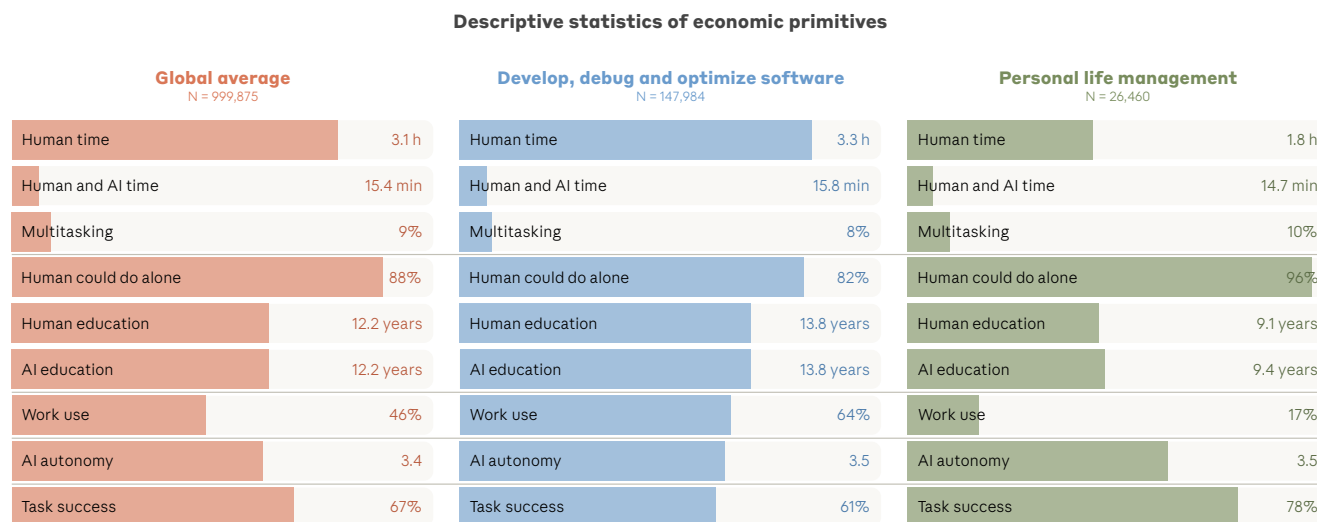


## Primitives highlight how use cases differ

To illustrate how the primitives distinguish between different types of AI use, we examine two contrasting request clusters: software development (“Help debug, develop, and optimize software across multiple programming domains”) and personal life management (“Assist with personal life management and everyday tasks”). Figure 2.2 shows the primitive profile for each cluster alongside global averages.



**Figure 2.2: Descriptive statistics of economic primitives overall, and for two example request clusters.**

For this figure, we focus on descriptive statistics for the primitives across the whole Claude.ai sample as well as two request clusters at the lowest level of granularity. N indicates the overall count of conversations or the count of conversations belonging to the request clusters.

**Task complexity.** Claude estimates that software development requests would take a competent professional approximately 3.3 hours to complete without AI—close to the global average of 3.1 hours. Personal life management tasks are estimated to be simpler, averaging 1.8 hours. Estimated human-AI collaboration time is similar across both (~15 minutes), showing this primitive varies less than other primitives for these two tasks.

**Human and AI skills.** Software development requests draw on more specialized knowledge: both human prompts and AI responses are estimated to require approximately 13.8 years of education to understand, compared to 9.1–9.4 years for personal life management requests. Claude estimates that users would be able to complete personal life management requests by themselves 96% of the time, versus 82% for software development requests—indicating that Claude provides more essential support for technical work.

**Use case.** Claude classifies 64% of software development requests as work-related, compared to just 17% for personal life management. This illustrates that Claude can be used for very different purposes. Overall, Claude.ai use is 46% work, 19% coursework, and 35% personal.

**AI autonomy.** Both clusters show similar estimated autonomy levels (~3.5 on a 1 to 5 scale), near the global average. This means that both software development and personal life management tasks, on average, afford Claude a similar autonomy to make decisions on how to complete the task.

**Task success.** Claude assesses personal tasks as successfully completed 78% of the time, versus 61% for software development. Harder tasks—those requiring more specialized knowledge and where users could not easily complete them alone—show lower estimated success rates.

## Tasks and primitives differ between Claude.ai and API users

As in our previous report, we find major differences in the tasks and primitives in Claude.ai conversations compared to the 1P API data. Part of this reflects the nature of the interaction: Claude.ai transcripts can include multi-turn conversations, while the API data we analyze is limited to single input-output pairs. This is because API requests arrive independently, with no metadata linking them to prior exchanges. This means we can only analyze them as isolated user-assistant pairs rather than full conversation trajectories.

Overall, API usage is overwhelmingly work-related (74% vs. 46%) and directive (64% vs. 32%), with three-quarters of interactions classified as automation compared to less than half on Claude.ai (see Figure 1.3).

Claude.ai users, by contrast, engage in more back-and-forth: task iteration and learning modes are far more common, and tasks tend to be more lengthy—both in terms of human time with AI (15 minutes vs. 5 minutes) and the estimated time a human would need to complete the task alone (3.1 hours vs. 1.7 hours). Claude.ai also shows higher task success rates (67% vs. 49%), which may reflect the benefits of multi-turn conversation, where users can clarify, correct course, and iterate toward a solution. Claude.ai users also give the AI

more autonomy on average, and are more likely to bring tasks they couldn't complete alone.

These differences are also reflected in the occupational distribution of tasks. API usage is heavily concentrated in Computer & Mathematical tasks (52% vs. 36%), consistent with its use for programmatic, automation-friendly workflows like code generation and data processing. Office & Administrative tasks are also more prevalent in the API (15% vs. 8%), reflecting routine business operations suited to delegation. Claude.ai, by contrast, sees substantially more Educational Instruction tasks (16% vs. 4%)—coursework help, tutoring, and instructional material development—as well as more Arts, Design, and Entertainment tasks (11% vs. 6%). Claude.ai also has a longer tail of human-facing categories like Community & Social Service and Healthcare Practitioners, where users seek advice, counseling, or information on personal matters.

These patterns suggest that 1P API deployments concentrate on tasks amenable to systematic automation, while Claude.ai serves a broader range of use cases including learning, creative work, and personal assistance.

Chapter 4 explores task-level variation in greater depth.

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<sup>1</sup> A classifier is a model that assigns a given input (e.g., a user conversation) a specific output (e.g., the use case “work”). In this report, we use Claude as a classifier, meaning that we prompt Claude to select a specific output and then use Claude’s response as the output (see Table 2.1 for the prompts).

<sup>2</sup> Throughout this report, we use binned scatterplots to show bivariate relationships. We divide observations into 20 equally-sized bins based on the x variable, then plot the average x and y values for each bin. The leftmost dot, for example, represents the averages for observations in the lowest 5% of the x distribution.

## CHAPTER 3

# How Claude is used varies by geography

## Overview

In this chapter, we analyze geographic variation in Claude usage patterns using a privacy-preserving<sup>1</sup> analysis of 1 million Claude.ai conversations.<sup>2</sup> We make five observations:

**Claude is mostly used for work, but use cases diversify with adoption:** Work and personal use cases are more common in higher-income countries, while coursework use cases are more common in lower-income countries. This echoes findings from our prior report and aligns with [recent work by Microsoft](#).

**GDP and human education predict adoption globally and within the US:** A 1% increase in GDP per capita is associated with a 0.7% increase in Claude usage per capita at the country level. Human education—Claude’s estimate of years of formal education needed to understand the human prompt—correlates positively with the Anthropic AI Usage Index at both levels.

**Other primitives predict adoption differently at global vs. US levels:** At the country level, higher usage correlates with shorter tasks and less AI autonomy. At the US state level, these relationships are not statistically significant, though work use correlates positively with adoption.

**Relationships between primitives depend on context:** Task success is negatively associated with human education across countries, but positively within US states. However, when controlling for other primitives, the US relationship becomes insignificant.

**How humans prompt is how Claude responds:** The education levels of human prompts and AI responses are nearly perfectly correlated ( $r > 0.92$  at both levels). Higher per capita usage countries also show more augmentation—using Claude as a collaborator rather than delegating decisions entirely.